



Chapter 23 – Project planning

Topics covered



- ✧ Software pricing
- ✧ Plan-driven development
- ✧ Agile planning

Project planning



- ✧ Project planning involves breaking down the work into parts and assign these to project team members, anticipate problems that might arise and prepare tentative solutions to those problems.
- ✧ The project plan, which is created at the start of a project, is used to communicate how the work will be done to the project team and customers, and to help assess progress on the project.



Software pricing

Software pricing



- ✧ Estimates are made to discover the cost, to the developer, of producing a software system.
 - You take into account, hardware, software, travel, training and effort costs.
- ✧ There is not a simple relationship between the development cost and the price charged to the customer.
- ✧ Broader organisational, economic, political and business considerations influence the price charged.

Factors affecting software pricing



Factor	Description
Contractual terms	A customer may be willing to allow the developer to retain ownership of the source code and reuse it in other projects. The price charged may then be less than if the software source code is handed over to the customer.
Cost estimate uncertainty	If an organization is unsure of its cost estimate, it may increase its price by a contingency over and above its normal profit.
Financial health	Developers in financial difficulty may lower their price to gain a contract. It is better to make a smaller than normal profit or break even than to go out of business. Cash flow is more important than profit in difficult economic times.

Factors affecting software pricing



Factor	Description
Market opportunity	A development organization may quote a low price because it wishes to move into a new segment of the software market. Accepting a low profit on one project may give the organization the opportunity to make a greater profit later. The experience gained may also help it develop new products.
Requirements volatility	If the requirements are likely to change, an organization may lower its price to win a contract. After the contract is awarded, high prices can be charged for changes to the requirements.

Pricing strategies



✧ Under pricing

- A company may underprice a system in order to gain a contract that allows them to retain staff for future opportunities
- A company may underprice a system to gain access to a new market area

✧ Increased pricing

- The price may be increased when a buyer wishes a fixed-price contract and so the seller increases the price to allow for unexpected risks

Pricing to win



- ✧ The software is priced according to what the software developer believes the buyer is willing to pay
- ✧ If this is less than the development costs, the software functionality may be reduced accordingly with a view to extra functionality being added in a later release
- ✧ Additional costs may be added as the requirements change and these may be priced at a higher level to make up the shortfall in the original price



Plan-driven development

Plan-driven development



- ✧ Plan-driven or plan-based development is an approach to software engineering where the development process is planned in detail.
 - Plan-driven development is based on engineering project management techniques and is the 'traditional' way of managing large software development projects.
- ✧ A project plan is created that records the work to be done, who will do it, the development schedule and the work products.
- ✧ Managers use the plan to support project decision making and as a way of measuring progress.

Project plans



✧ In a plan-driven development project, a project plan sets out the resources available to the project, the work breakdown and a schedule for carrying out the work.

✧ Plan sections

- Introduction
- Project organization
- Risk analysis
- Hardware and software resource requirements
- Work breakdown
- Project schedule
- Monitoring and reporting mechanisms

Project plan supplements



Plan	Description
Configuration management plan	Describes the configuration management procedures and structures to be used.
Deployment plan	Describes how the software and associated hardware (if required) will be deployed in the customer's environment. This should include a plan for migrating data from existing systems.
Maintenance plan	Predicts the maintenance requirements, costs, and effort.
Quality plan	Describes the quality procedures and standards that will be used in a project.
Validation plan	Describes the approach, resources, and schedule used for system validation.

The planning process



- ✧ Project planning is an iterative process that starts when you create an initial project plan during the project startup phase.
- ✧ Plan changes are inevitable.
 - As more information about the system and the project team becomes available during the project, you should regularly revise the plan to reflect requirements, schedule and risk changes.
 - Changing business goals also leads to changes in project plans. As business goals change, this could affect all projects, which may then have to be re-planned.

Planning assumptions



- ✧ You should make realistic rather than optimistic assumptions when you are defining a project plan.
- ✧ Problems of some description always arise during a project, and these lead to project delays.
- ✧ Your initial assumptions and scheduling should therefore take unexpected problems into account.
- ✧ You should include contingency in your plan so that if things go wrong, then your delivery schedule is not seriously disrupted.

Risk mitigation



- ✧ If there are serious problems with the development work that are likely to lead to significant delays, you need to initiate risk mitigation actions to reduce the risks of project failure.
- ✧ In conjunction with these actions, you also have to re-plan the project.
- ✧ This may involve renegotiating the project constraints and deliverables with the customer. A new schedule of when work should be completed also has to be established and agreed with the customer.



Agile planning

Agile planning



- ✧ Agile methods of software development are iterative approaches where the software is developed and delivered to customers in increments.
- ✧ Unlike plan-driven approaches, the functionality of these increments is not planned in advance but is decided during the development.
 - The decision on what to include in an increment depends on progress and on the customer's priorities.
- ✧ The customer's priorities and requirements change so it makes sense to have a flexible plan that can accommodate these changes.

Agile planning stages



- ✧ Release planning, which looks ahead for several months and decides on the features that should be included in a release of a system.
- ✧ Iteration planning, which has a shorter term outlook, and focuses on planning the next increment of a system. This is typically 2-4 weeks of work for the team.

Approaches to agile planning



✧ Planning in Scrum

- Covered in Chapter 3

✧ Based on managing a project backlog (things to be done) with daily reviews of progress and problems

✧ The planning game

- Developed originally as part of Extreme Programming (XP)
- Dependent on user stories as a measure of progress in the project

The planning game



Agile planning applicability



- ✧ Agile planning works well with small, stable development teams that can get together and discuss the stories to be implemented.
- ✧ However, where teams are large and/or geographically distributed, or when team membership changes frequently, it is practically impossible for everyone to be involved in the collaborative planning that is essential for agile project management.